

MICHAEL G. RIVERA ARROYO

Crew Chief · Instrumentation Specialist · Systems Builder with AI

Trujillo Alto / San Juan, Puerto Rico

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GitHub: github.com/mgriveraarroyo

Puerto Rico-based · Bilingual Spanish / English · On-site, hybrid, or remote



Scan: GitHub

PROFESSIONAL SUMMARY

Field operations professional with 5+ years leading survey crews and subsurface utility studies using Trimble GPS, Total Station, and GPR technologies (US Radar, GSSI). Self-taught Systems Builder with two production client websites deployed, a personal AI operating system (more than 50 custom tools), and hands-on integration with Anthropic, OpenAI, and local LLM stacks. Combines field-grade precision with modern software engineering and multi-agent AI orchestration.

TECHNICAL SKILLS

AI / ML: Anthropic SDK · OpenAI SDK · llama.cpp · LanceDB · ONNX runtime · Playwright (authenticated automation) · prompt engineering · multi-agent orchestration · RAG pipelines · ElevenLabs TTS.

Languages: Python · Bash · JavaScript · HTML / CSS · YAML · Markdown · JSONL.

Infrastructure & Tools: Git · GitHub Pages · GitHub API · macOS launchd / LaunchAgents · webhook RSA signing · Cloudflared tunnels · Square integration · Instagram Graph API v21 · cron · jsonl audit logs.

Field / Survey: Trimble GPS · Trimble Total Station · Ditch Witch · Vermeer G3 · Ridgid · US Radar GPR · GSSI GPR · GIS · Blueprint reading · Crew leadership · Field safety.

Methodology: Specification-first development · iterative QA with external validation · invariants-first architecture · disciplined Git workflow (no force-push, signed commits).

PROFESSIONAL EXPERIENCE

Founder · Systems Builder (Self-employed)

Trujillo Alto, PR · 2026 – Present

End-to-end design and operation of a personal AI operating system and a client website pipeline. Selected achievements:

- Built **2 production client websites** deployed on GitHub Pages (24/7 uptime, HTTP 200 verified, SHA-matched deploys), with 91.4% external QA pass rate on the most-audited site.
- Authored **more than 50 custom command-line tools** (Bash + Python) covering automation, monitoring, RAG search, voice synthesis, browser automation, and external API bridges.
- Architected **22 active macOS LaunchAgents** for backup, monitoring, nightly RAG re-index, and continuous health checks.
- Implemented **local-first AI stack** (llama.cpp + LanceDB + nomic-embed + ONNX) running entirely on local hardware — reduced cloud API spend to \$0 on daily research workloads.
- Engineered **authenticated browser automation** (Playwright) for Instagram / Facebook / X data extraction; fixed real bugs in session state handling and SPA settle logic, verified output 1:1 against ground-truth screenshots.
- Developed **R&D crypto trading bot** (paper-only) implementing HMM regime detection, fee-aware backtests, and Kronos inference on Coinbase OHLCV data.
- Established an **operational framework** with 8 documented best practices, 10 core architecture principles, and an 11-rule project playbook built from real client deployment cycles.

Crew Chief / Instrumentation Specialist

Javier E. Bidot Associates, PSC (Land Surveyors & Consultants) · 2019 – 2025

Assignments spanned engineering contractors including Arcadis and DPA, with field deployments across Puerto Rico and Florida.

- **Thousands of cumulative field hours** operating Trimble GPS, Trimble Total Station, and a full EM/GPR locator suite (Ditch Witch, Vermeer G3, Ridgid, US Radar GPR, GSSI GPR) — documented in company timesheets across 5+ years of daily deployment.
- **Project types delivered:** topographic surveys, subsurface utility studies (potable water and others), EM locating, GPR scanning, and blueprint-based field layout.

- **Daily field responsibilities:** instrument operation and calibration, data capture accuracy, blueprint interpretation, GIS-ready deliverable preparation, and on-site coordination of crew tasks.
- **On-the-job training** under licensed surveyors of the PSC firm, with strict safety protocols and bilingual coordination with US-based contractors and PR-based agencies.

Construction Assistant

Inteira Corporation · San Juan, PR · 2017 – 2019

- Supported foreman on jobsite layout, measurements, and project coordination.
- Managed materials, tools, and purchase orders for active sites.

Hospitality Operations

San Juan, PR · 2008 – 2017

Nine years in high-volume hospitality (cooking, service, and operations · including airport restaurants). Transferable skills: sustained-pressure performance, team coordination, process discipline.

PROJECTS

- **Strands & Roses (Atlanta-based hair/lash/fine-art studio site)** · live 24/7 · <https://mgriveraarroyo.github.io/strandsandroses/> · static HTML/CSS/JS · key-free architecture · Square booking, IG feed, and Google Maps integration · device-aware responsive design · 91.4% external QA pass.
- **Richard / D.E.L.O. (Editorial music & brand site)** · live 24/7 · <https://mgriveraarroyo.github.io/richard/> · custom SVG monogram, embedded Spotify, mobile 320px verified in real-browser QA, honest-visible form failure states.
- **TradingBot v2 (Research)** · ETH paper trading framework with HMM regime detection (v3 smoothed), Kronos inference, fee-aware backtests, and triple-gate risk management.
- **FORGE OS (Personal AI operating system)** · more than 50 custom tools, 22 LaunchAgents, local-first AI stack, multi-agent orchestration with Anthropic + OpenAI + local llama.cpp.

EDUCATION & CERTIFICATIONS

- **Instituto de Banca**, Hato Rey, PR · *Cocina Local e Internacional* · 2009 – 2010.
- **Self-directed continuing education** (2026 – Present) · software engineering, AI orchestration, browser automation, trading systems R&D · evidence via GitHub portfolio.

References available upon request.